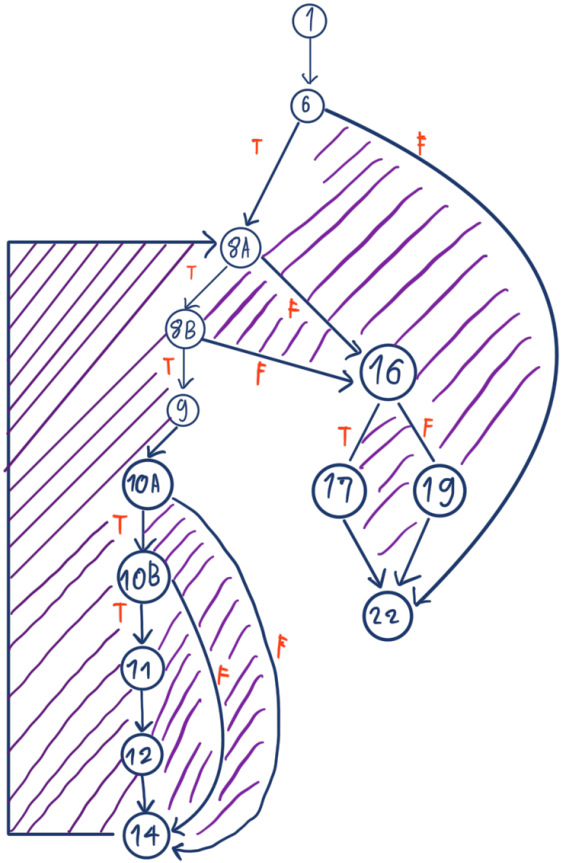
```
var average = function (value, min, max) {
    let total = new Object(), average = 0;
    total.valid = 0;
    total.input = 0;
    //check whether value is an array
    if (Array.isArray(value)) {
        let i = 0, sum = 0;
        while (value[i] !== -999 && total.input < 100) {
            (total.input)++;
            if (value[i] >= min && value[i] <= max) {
                (total.valid)++;
                sum = sum + value[i];
            }
            i++;
        }
        if (total.valid > 0) {
            average = sum / total.valid;
        } else {
            average = -999;
        }
    }
    return {total:total,average:average };
```



(1) (start)
(6) (check value ว่าเป็น array มั้ย)
(8A) (While (value[i] !== -999))
(8B) (While (total.input < 100))
(9) (total. input ++)
(10A) (if (value[i] >= min)
(10B) (if (value[i] <= max)
(11) (total.valid ++)
(12) (sum += value)
(14) (เพิ่มค่า i++ ใน array)
(16) (check total.valid > 0 หรือไม่)
(17) (คำนวณ average = sum / total.valid)
(19) (กำหนดค่า average = -999
(22) (คืนค่า total, average)

ข้อที่ 2

1) # Regions V(G) = 7
2) (6) (check value ว่าเป็น array มั้ย)
   (8A) (While (value[i] != -999))
   (8B) (While (total.input < 100))
   (10A) (if (value[i] >= min)
   (10B) (if (value[i] <= max)
   (16) (check total.valid > 0 หรือไม่)

   Cyclomatic Complexity = 6+1 = 7

3) V(G) = E - N + 2
        = 19 - 14 + 2
        = 7

ข้อที่ 3-4

Basis Path

1) 1 → 6 → 8A → 8B → 9 → 10A → 10B → 11 → 12 → 14
2) 1 → 6 → 8A → 8B → 16 → 19 → 22
3) 1 → 6 → 8A → 16 → 17 → 22
4) 1 → 6 → 8A → 16 → 19 → 22
5) 1 → 6 → 8A → 8B → 9 → 10A → 14
6) 1 → 6 → 8A → 8B → 9 → 10A → 10B → 14
7) 1 → 6 → 22

Basis Path	Test Case ID	Test objective	Test data	Expected Output	Actual Output	Test Result (Passed หรือ Failed)
1→6→8A→8B→ วนลูปหลายรอบ→ 16→19→22	Testcase 1	ทดสอบค่าทั้งหมดที่อยู่ในช่วง min - max	Value = [10, 20, 30], min = 5, max = 50	average = 20 total.valid = 3	average = 20 total.valid = 3	Passed
1→6→8A→8B→9→ 10A→10B→12→ 8A→16→17→22	Testcase 2	ทดสอบกรณีที่มีบางค่า หลุดออกนอกช่วงเกณฑ์	Value=[10,200,30], min = 5, max = 50	average = 20 total.valid = 2	average = 20 total.valid = 2	Passed
1→6→8A→ 8B→9→14→ 8A→16→17→22	Testcase 3	ทดสอบกรณีที่ค่าทั้งหมดหลุดออกนอกช่วงเกณฑ์	Value = [200, 300, 400], min = 5, max = 50	average = -999 total.valid = 0	average = -999 total.valid = 0	Passed
1→22	Testcase 4	ทดสอบกรณีที่ input ไม่ใช่ array	Value = "string", min = 5, max = 50	average = -999 total.valid = 0	average = 0 total.valid = 0	Failed
1→6→22	Testcase 5	ทดสอบกรณีที่ array [] ไม่มีค่า	Value = [], min = 5, max = 50	average = -999 total.valid = 0	average = -999 total.valid = 0	Passed